

Project: FitCheck

AI-Powered Fashion E-Commerce Platform

1. Project Title and Short Introduction

Project Title:

FitCheck — AI-Powered Multi-Brand Fashion E-Commerce Platform

Problem Statement

Online fashion shoppers globally face two critical pain points: fragmentation and fit uncertainty. Shoppers must visit 5–10 different brand websites to compare styles and prices, causing decision fatigue and abandoned carts. Additionally, 30% of all online clothing purchases are returned due to fit mismatch — costing brands billions annually in reverse logistics. Model photos fail to represent diverse body types, further eroding buyer confidence and driving up return rates.

Our Solution

FitCheck is a wearable technology startup that designs AI-powered smartwatches capable of continuously monitoring vital signs — heart rate, SpO2 (blood oxygen), body temperature, sleep patterns, and activity levels. Using on-device and cloud-based machine learning models, the watch proactively detects health anomalies, delivers personalized health coaching, and automatically triggers emergency SOS calls when critical thresholds are breached. A companion mobile application and web dashboard give users, caregivers, and healthcare providers a complete view of health trends and AI-generated recommendations.

Key Differentiators

- AI Virtual Try-On that renders clothing photorealistically on the shopper's own uploaded photo
- Smart Size Matching using body measurements and brand-specific sizing data to recommend perfect fits
- Unified multi-brand marketplace with 100+ fashion brands searchable by price, size, and style
- Brand analytics dashboard providing real-time sizing trends and try-on conversion data
- Freemium model with basic free access and unlimited try-ons for premium members

2. Business Model Canvas

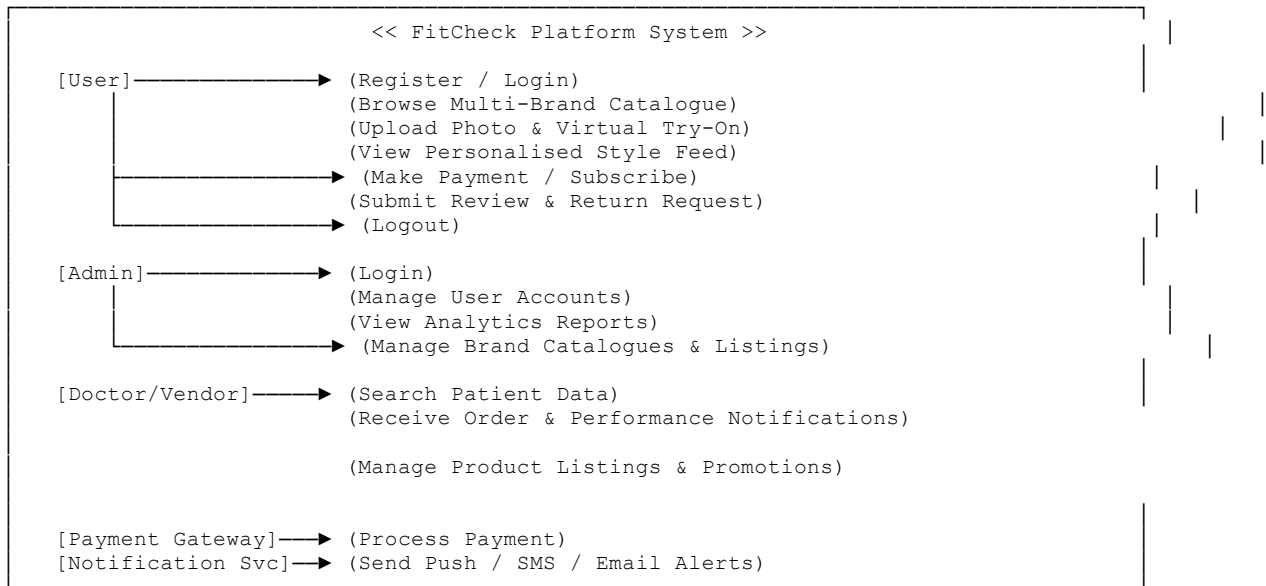
The Business Model Canvas below outlines all nine strategic building blocks for FitCheck:

KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITIONS	CUSTOMER RELATIONSHIPS	CUSTOMER SEGMENTS
• Chip/sensor manufacturers (Qualcomm, Sony)• Health data labs• Telecom providers (5G SIM)• App store platforms• Insurance companies	• AI model training & updates• Watch hardware R&D• App ecosystem development• Cloud infrastructure ops• Customer support & warranty	• Real-time AI health monitoring• Predictive disease alerts• Personalized fitness coaching• Emergency SOS auto-call• Seamless smartphone integration• 7-day battery life	• AI-powered chatbot support• Push notifications & alerts• Community health challenges• Subscription premium care• In-app loyalty rewards	• Health-conscious adults (25–55)• Chronic disease patients• Athletes & fitness enthusiasts• Elderly users (fall detection)• Corporate wellness programs
KEY RESOURCES	CHANNELS	REVENUE STREAMS		
		• Device sales (\$199–\$499)• Premium subscription (\$9.99/mo)• Health insurance partnerships• Data analytics licensing• Corporate wellness packages		
• Proprietary AI health algorithms• Hardware patents• Cloud data infrastructure• Skilled AI/ML engineers• Brand & community	• Official website & app store• Retail stores (Best Buy, Carrefour)• B2B corporate sales• Healthcare clinics• Social media & influencers	<i>(See revenue streams above)</i>		
COST STRUCTURE	• Hardware manufacturing & procurement • AI/ML R&D and model training • Cloud server hosting & maintenance • Sales, marketing & distribution • Customer support operations			

3. Use Case Diagram

The following diagram illustrates how different actors interact with the Smart AI Watch system. Actors include the end User, Admin, Doctor/Vendor, Payment Gateway, and Notification Service.

USE CASE DIAGRAM — Smart AI Watch System



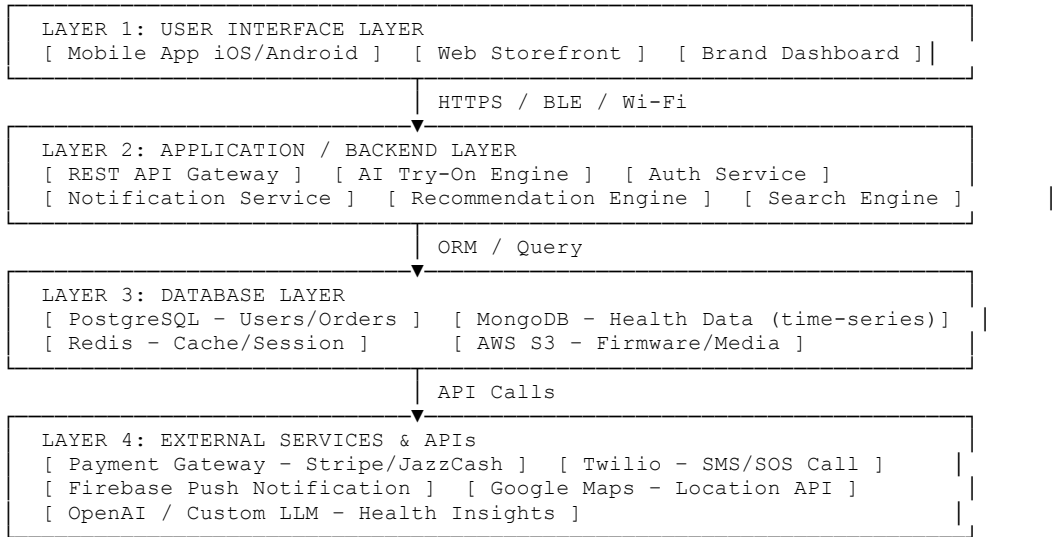
Actor Descriptions

- User: Registers, browses catalogue, uploads photo for virtual try-on, makes purchases, manages orders
- Admin: Manages user and brand accounts, monitors platform health, approves new brands, views analytics reports
- Brand/Vendor: Manages product listings, accesses customer analytics, receives order and performance notifications
- Payment Gateway: Processes product purchases and subscription payments (Stripe, JazzCash, credit/debit cards)
- Notification Service: Delivers push notifications, order updates, and promotional alerts (Firebase, Twilio)

4. System Architecture Diagram

The Smart AI Watch system follows a four-layer architecture: User Interface, Application Backend, Database, and External Services. Each layer communicates through secure HTTPS APIs and BLE (Bluetooth Low Energy) for watch-to-phone communication.

SYSTEM ARCHITECTURE DIAGRAM — FitCheck Platform



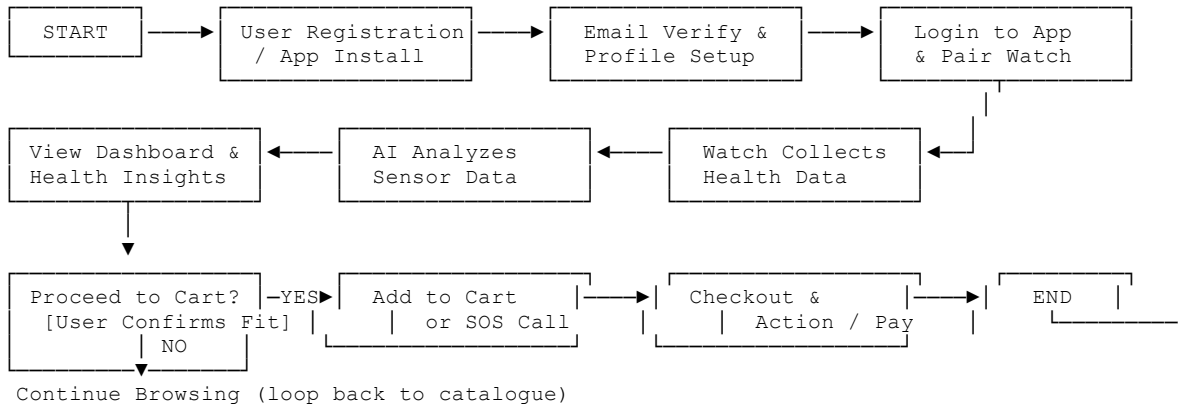
Layer Descriptions

- **UI Layer:** iOS/Android mobile app, web storefront for shoppers, and brand analytics web dashboard
- **Backend Layer:** REST API Gateway, AI Try-On Engine (computer vision + generative AI), Auth Service (JWT), Recommendation & Search Engine
- **Database Layer:** PostgreSQL for relational user/order data, MongoDB for product catalogue and try-on sessions, Redis for session caching, AWS S3 for images and media
- **External Services:** Stripe/JazzCash (payments), Twilio (order SMS alerts), Firebase (push notifications), Google Maps (delivery tracking)

5. Workflow Diagram

The workflow diagram below shows the complete step-by-step operational flow of the Smart AI Watch system from user onboarding through continuous health monitoring and alert handling.

WORKFLOW DIAGRAM — FitCheck Platform System



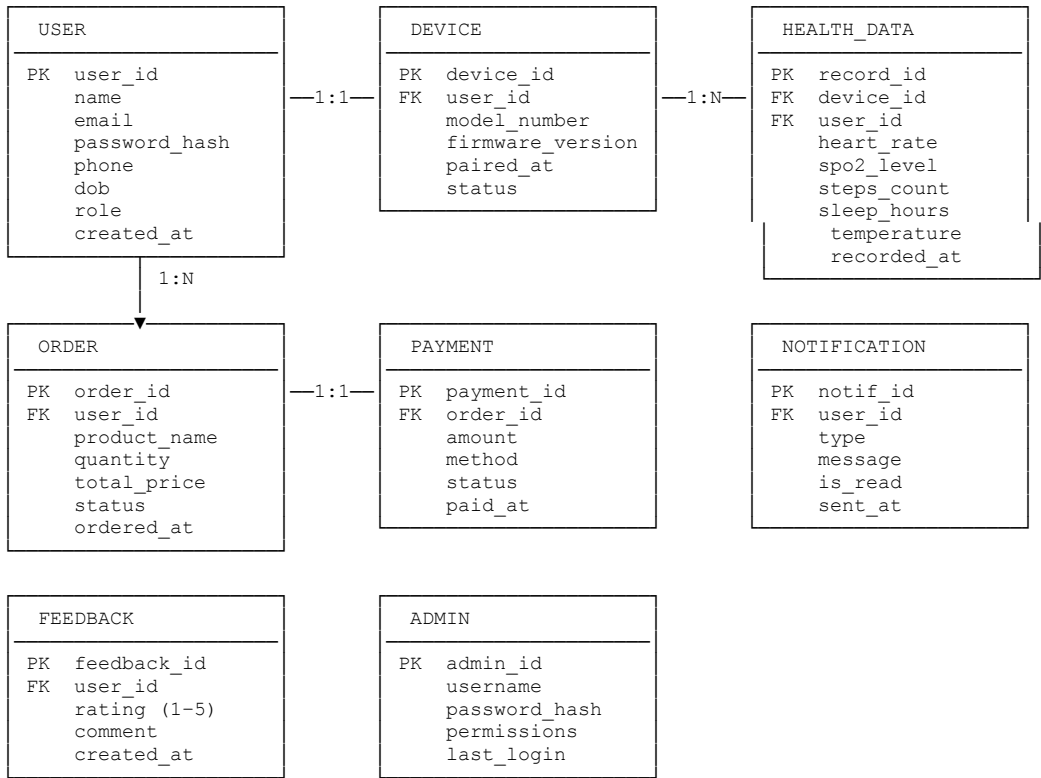
Workflow Steps Explained

- Step 1 — Registration: User downloads app, creates account, verifies email, and sets up style profile (sizes, brand preferences, body measurements)
- Step 2 — Catalogue Browsing: User browses 100+ brands filtered by price, size, category, and style preferences
- Step 3 — Photo Upload & Item Selection: User uploads a full-body photo and selects any item to virtually try on
- Step 4 — AI Try-On Rendering: Cloud AI engine processes the photo and renders the selected item onto the user's body with photorealistic accuracy
- Step 5 — Fit Decision: User reviews the virtual try-on result and receives AI-generated size and fit recommendations
- Step 6 — Purchase & Checkout: User adds item to cart and completes secure payment via Stripe or JazzCash
- Step 7 — Order Fulfilment: Order is routed to the brand/partner for dispatch; user receives real-time delivery tracking notifications

6. Entity Relationship Diagram (ERD)

The ERD below shows the complete database design for the Smart AI Watch system, including all entities, attributes, primary keys (PK), foreign keys (FK), and relationships.

ENTITY RELATIONSHIP DIAGRAM (ERD) — FitCheck Platform



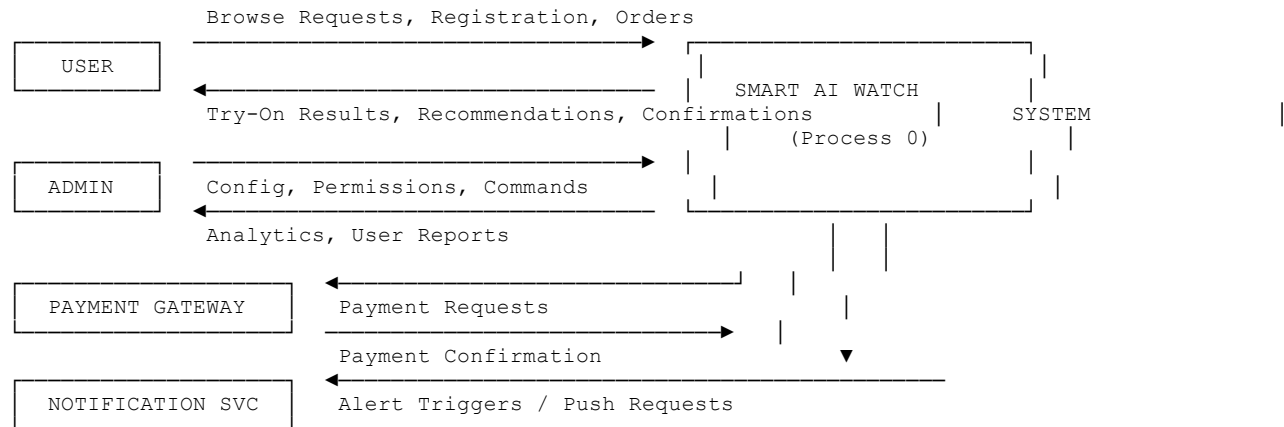
Relationship Summary

- **USER** —1:1—► **DEVICE** (each user has one paired watch)
- **USER** —1:N—► **HEALTH_DATA** (user generates many health records via device)
- **USER** —1:N—► **ORDER** (user can place multiple orders)
- **ORDER** —1:1—► **PAYMENT** (each order has one payment record)
- **USER** —1:N—► **NOTIFICATION** (user receives many notifications)
- **USER** —1:N—► **FEEDBACK** (user submits feedback on service)

7. DFD Level 0 — Context Diagram

The context diagram shows the Smart AI Watch System as a single high-level process (Process 0), surrounded by all external entities and the flow of data between them.

DFD LEVEL 0 — Context Diagram



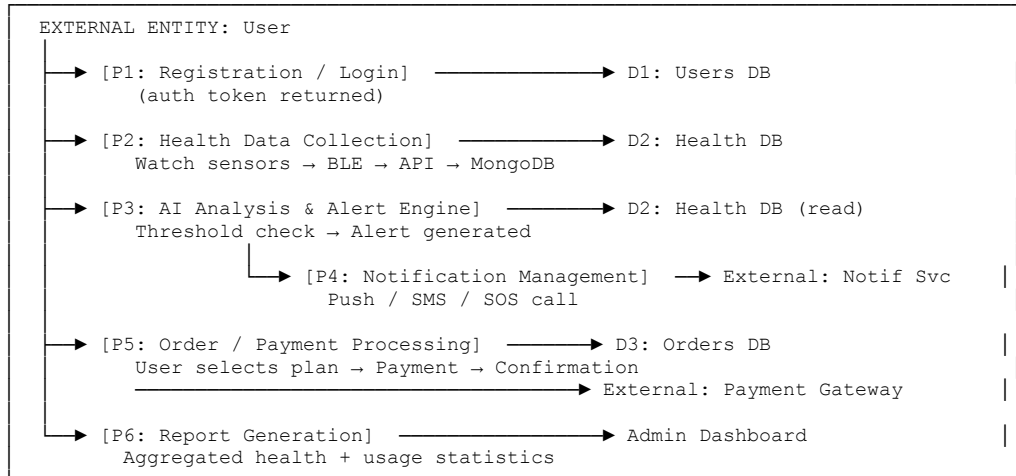
Data Flow Summary

- User → System: Registration details, health sensor data, order requests, payment info
- System → User: Health alerts, AI insights, order confirmations, reports
- Admin → System: Management commands, configuration, access control
- System → Admin: Analytics dashboards, user reports, system health data
- System → Payment Gateway: Payment authorization requests
- System → Notification Service: Alert triggers, SOS requests

8. DFD Level 1

DFD Level 1 expands the central process into six functional sub-processes, showing how data flows between modules, data stores, and external entities.

DFD LEVEL 1 — System Breakdown



Process Descriptions

- P1 — Registration/Login: Handles user onboarding, authentication, JWT token issuance, and style profile setup. Data stored in D1: Users DB
- P2 — Virtual Try-On Processing: Receives user photo and selected item, processes through AI computer vision engine, stores try-on sessions in D2: Try-On DB (MongoDB)
- P3 — AI Recommendation Engine: Reads try-on history and browsing data, runs personalisation models, generates style recommendations and size suggestions
- P4 — Notification Management: Receives order and alert signals from P3/P5 and dispatches push notifications, order updates, and promotional alerts via Firebase/Twilio
- P5 — Order/Payment Processing: Manages product purchases, brand commissions, and subscription plans; integrates with Payment Gateway; stores records in D3: Orders DB
- P6 — Report Generation: Aggregates sales trends, try-on conversion rates, and sizing data for brand dashboards and admin analytics panels

9. Activity Diagram — Health Alert Flow (Bonus)

This bonus activity diagram illustrates the detailed decision flow for the AI health alert pipeline — from sensor reading through to emergency response and database logging.

ACTIVITY DIAGRAM — Virtual Try-On & Purchase Flow (Bonus)

